

CHAPTER 6

Attitude dynamics

6.1. Space story

We were flying a spacecraft in transfer orbit that had a liquid apogee engine (LAE). This meant that unlike our other spacecraft it had an oxidizer tank. Our particular LAE engines used hydrazine, so the same propellant could be used in the monopropellant engines and the LAE. We were spinning at 10 rpm for our first LAE maneuver. When it started up we noticed a slight oscillation that damped quickly. Nutation was damped by the nutation controllers that had been designed to correct nutation at tip-off when the tanks were full. As speed was important at the tip-off it did not have any roll-off. This made sense because there were no frequencies greater than nutation when the tanks were full.

On the next burn, we lowered the spin rate to reduce fuel consumption. We started the burn and the control system started limit-cycling. Control was stable so I did not stop the burn. We had an extra year of fuel so a little wasted during an LAE burn was not important. As the customer wanted to get to mission orbit fast, this was a reasonable decision. Management did not agree but the mission went on. The problem was that as the tanks emptied slosh motion of the fuel was greater and it was worse at the lower frequency. We spun up to 10 rpm and the problem went away.

6.2. Introduction

This chapter provides an introduction to spacecraft-attitude dynamics. In this chapter, we will discuss rigid-body rotation. This motion is in general, nonlinear and all three axes are coupled. The following sections cover multibody dynamics. The models presented are central-hub models. We first discuss the gyrostator that is a rigid body with any number of symmetric rotors. The gyrostator is important for spacecraft with reaction wheels. We then cover models that allow for more than one degree of rotational freedom at the joint and include the effects of center-of-mass movement. Finally, we discuss a spacecraft in which part of the spacecraft can undergo infinitesimal displacements. This is representative of a spacecraft with an attached flexible structure.

The approach taken in most cases in this chapter is to derive the equations of motion based on momentum conservation. The planar slosh problem uses Lagrange's method. Another well-known method is Kane's method that involves generalized accelerations [1]. Many general-purpose computer codes exist for modeling complex multibody systems, such as topological trees with and without closed loops. Some software packages use symbolic derivation to produce compact and efficient multibody models.

6.3. Inertia matrix

6.3.1 Definition

The inertia matrix represents the resistance to the rotation of a spacecraft. The inertia matrix is positive-definite-symmetric, which means that the direction of rotation does not matter. The nine elements are

$$I_{xx} = \int \int \int_V \rho(x, y, z)(y^2 + z^2) \, dx dy dz \quad (6.1)$$

$$I_{yy} = \int \int \int_V \rho(x, y, z)(x^2 + z^2) \, dx dy dz \quad (6.2)$$

$$I_{zz} = \int \int \int_V \rho(x, y, z)(x^2 + y^2) \, dx dy dz \quad (6.3)$$

$$I_{xy} = - \int \int \int_V \rho(x, y, z)xy \, dx dy dz \quad (6.4)$$

$$I_{xz} = - \int \int \int_V \rho(x, y, z)xz \, dx dy dz \quad (6.5)$$

$$I_{yz} = - \int \int \int_V \rho(x, y, z)yz \, dx dy dz \quad (6.6)$$

where the vectors are from the center-of-mass. Positive-definite means that

$$\mathbf{z}^T \mathbf{I} \mathbf{z} > 0 \quad (6.7)$$

for all nonzero vectors $\mathbf{z}$ with real entries. All inertia matrices can be transformed, through the appropriate change in axes, into a diagonal matrix.

The center-of-mass is found from

$$c_x = \frac{1}{M} \int \int \int_V \rho(x, y, z)x \, dx dy dz \quad (6.8)$$

$$c_y = \frac{1}{M} \int \int \int_V \rho(x, y, z)y \, dx dy dz \quad (6.9)$$

$$c_z = \frac{1}{M} \int \int \int_V \rho(x, y, z)z \, dx dy dz \quad (6.10)$$

where $\mathbf{c}$ is the vector to the center-of-mass and

$$M = \int \int \int_V \rho(x, y, z) \, dx dy dz \quad (6.11)$$

6.3.2 Inertia matrix from components

The diagonal terms must all be positive. $I_{xy} = I_{yx}$, $I_{xz} = I_{zx}$, and $I_{yz} = I_{zy}$. It is often useful to decompose a spacecraft into a set of point masses. The inertia matrix for N

point masses is

$$I = - \sum_{k=1}^N m_k r_k^\times r_k^\times \quad (6.12)$$

where

$$r_k^\times = \begin{bmatrix} 0 & -z_k & y_k \\ z_k & 0 & -x_k \\ -y_k & x_k & 0 \end{bmatrix} \quad (6.13)$$

and the contribution of one point mass is

$$I_k^\times = m_k \begin{bmatrix} y_k^2 + z_k^2 & -x_k y_k & -x_k z_k \\ -x_k y_k & x_k^2 + z_k^2 & -y_k z_k \\ -x_k z_k & -y_k z_k & x_k^2 + y_k^2 \end{bmatrix} \quad (6.14)$$

For point masses

$$c = \frac{\sum_{k=1}^N m_k r_k}{\sum_{k=1}^N m_k} \quad (6.15)$$

The point-mass approach is very useful when individual inertias for components are not available. This method can be expanded to include rigid bodies instead of point masses,

$$I = \sum_{k=1}^N (B_k I_k B_k^T - m_k r_k^\times r_k^\times) \quad (6.16)$$

where B_k transforms from the frame in which the inertia matrix is calculated to the spacecraft body frame.

6.3.3 Common inertia matrices

Table 6.1 gives inertia matrices for some common geometric solids. m is the mass, r is the radius for the sphere and cylinder, and l is the height for a cylinder. x , y , and z are box dimensions. The cylinder is aligned with the z -axis. Example 6.1 computes the inertia of a satellite composed of three boxes. Two represent the solar panels and one the bus. This is a reasonable first cut at the inertia of the satellite. The deployment struts are ignored because they have low masses. The satellite bus is treated as a solid since it is full of all the bus components. The model is shown in Fig. 6.1.

Table 6.1 Common inertia matrices.

Type	Inertia matrix
Solid Box	$\frac{m}{12} \begin{bmatrix} y^2 + z^2 & 0 & 0 \\ 0 & x^2 + z^2 & 0 \\ 0 & 0 & x^2 + y^2 \end{bmatrix}$
Solid cylinder	$\frac{m}{12} \begin{bmatrix} 3r^2 + l^2 & 0 & 0 \\ 0 & 3r^2 + l^2 & 0 \\ 0 & 0 & 6r^2 \end{bmatrix}$
Solid sphere	$\frac{2mr^2}{5} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

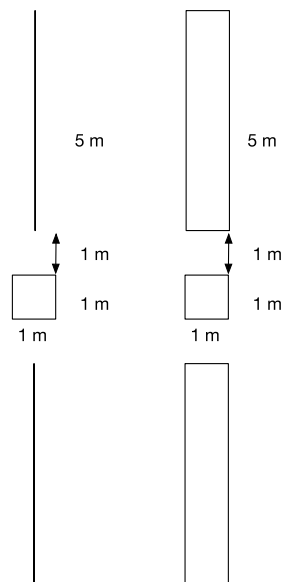


Figure 6.1 A satellite represented by three boxes.

6.4. Rigid body

The dynamical equations for a spacecraft can be derived by first writing the total angular momentum for the system in the inertial frame.

$$H = Ah \tag{6.17}$$

where h is the angular momentum in the body frame and the transformation matrix A transforms from the body-fixed coordinate frame to the inertial reference frame. The

```

1 %% Inertia calculation
2
3 % Solar panel
4 x = 1;
5 y = 5;
6 z = 0.01;
7 mSP = 2;
8 rN = [0;y/2+x;0];
9 rS = [0;-y/2-x;0];
10 iSP = (mSP/12)*diag([y^2+z^2 x^2+z^2 x^2+y^2])
11
12 % Bus
13 x = 1;
14 y = 1;
15 z = 1;
16 mBus = 10;
17 iBus = (mBus/12)*diag([y^2+z^2 x^2+z^2 x^2+y^2])
18
19 iTotal = iBus + 2*iSP - mSP*(SkewSq(rN) + SkewSq(
    rS))

```

```

1 iSP =
2
3     4.1667         0         0
4         0     0.1667         0
5         0         0     4.3333
6
7
8 iBus =
9
10     1.6667         0         0
11         0     1.6667         0
12         0         0     1.6667
13
14
15 iTotal =
16
17    59.0000         0         0
18         0     2.0000         0
19         0         0    59.3333

```

Example 6.1: Inertia matrix for a satellite represented by three boxes.

torque on the spacecraft is

$$\begin{aligned}
 AT = \dot{H} &= \dot{A}h + A\dot{h} \\
 \dot{A} &= A\omega^\times \\
 T &= \dot{h} + \omega^\times h
 \end{aligned} \tag{6.18}$$

where T is defined in the body frame and ω is the angular rate of the body with respect to the inertial frame measured in the body frame. For a rigid body $h = I\omega$ so

$$T = I\dot{\omega} + \omega^\times I\omega \tag{6.19}$$

which is Euler's equation. I does not have a time derivative when the spacecraft is rigid. The inertia matrix, I , is positive-definite-symmetric. This fact is important when we solve for the angular acceleration by solving the set of linear equations

$$I\dot{\omega} = T - \omega^\times I\omega \tag{6.20}$$

These equations are written about the spacecraft's center-of-mass. This decouples the angular motion from the translational motion.

Generally, we do not compute the inverse of I , rather we solve the set of linear equations for $\dot{\omega}$. As I is positive-definite-symmetric, we can solve this set of equations using a linear equation solver without pivoting, thus saving much computation. For a 3-by-3 system with a constant inertia matrix, we can precompute the inverse and not go through the process of using a solver. Pivoting is the process by which we find an element in a column with the largest absolute value and then do row shifts so that this number is in the row being considered.

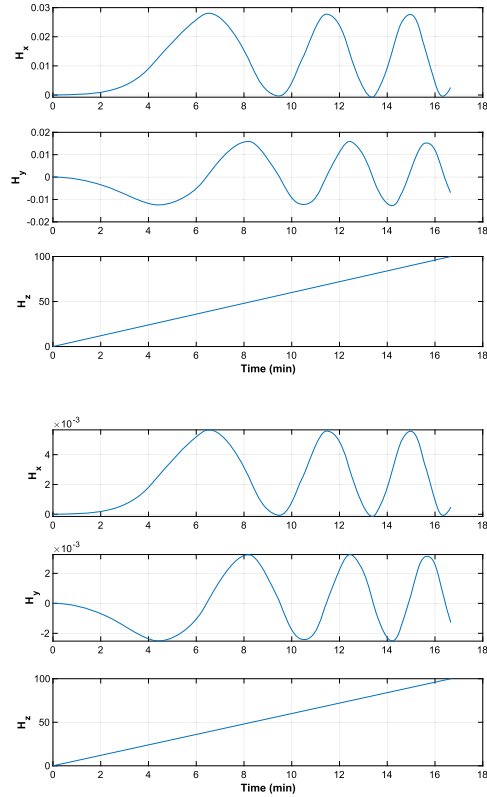
A is a transformation matrix that we normally represent as a quaternion. Quaternions are discussed in Section 5.5. To demonstrate rigid-body dynamics define an inertially fixed torque about the z -axis and a fully populated inertia matrix as in Example 6.2.

The first four elements of the state vector, x , is the quaternion that transforms from the inertial frame to the body frame with a starting value of $[1;0;0;0]$. The last three elements are the angular rate in the body frame that is $[0;0;0]$. The equations of motion are in the function FRB and the function RK4 is the fourth-order Runge–Kutta integrator. QTForm uses the quaternion to transform from the body frame to the inertial frame. QForm transforms from the inertial frame to the body frame.

```

1 inertia = [2000 300 -200;...
2           300 4000 -100;...
3           -200 -100 1000];
4 tEnd     = 1000;
5 dT       = 1.0;
6 n        = tEnd/dT;
7 torque   = [0;0;0.1];
8 xP       = zeros(3,n);
9 x        = [1;0;0;0;0;0;0];
10
11 for k = 1:n
12     xP(:,k) = QTForm(x(1:4),inertia*x(5:7));
13     tBody    = QForm( x(1:4), torque );
14     x = RK4( @RHS, x, dT, 0, inertia, tBody );
15 end
16
17 yL = {'H_x' 'H_y' 'H_z'};
18 s = sprintf('RigidBody_dT=%1.1f s',dT);
19 TimeHistory( (0:(n-1))*dT, xP, yL, s )
20
21 dT = 0.2;
22 n = tEnd/dT;
23 xP = zeros(3,n);
24 x = [1;0;0;0;0;0;0];
25
26 for k = 1:n
27     xP(:,k) = QTForm(x(1:4),inertia*x(5:7));
28     tBody    = QForm( x(1:4), torque );
29     x = RK4( @RHS, x, dT, 0, inertia, tBody );
30 end
31
32 s = sprintf('RigidBody_dT=%1.1f s',dT);
33 TimeHistory( (0:(n-1))*dT, xP, yL, s )
34
35 %% Right-hand-side for a rigid body
36 function xDot = RHS( x, ~, inertia, torque )
37
38 q = x(1:4);
39 omega = x(5:7);
40 qDot = QIToBDot(q,omega);
41 omegaDot = inertia \...
42     (torque - Cross(omega, inertia*omega));
43
44 xDot = [qDot; omegaDot];
45
46 end

```



Example 6.2: Rigid-body dynamics showing momentum conservation. Note the Y-axis scale. The top simulation has a time step of 1 second, the bottom has a time step of 0.2 seconds.

The top plot shows the simulation with a time step of 0.2 s. The z -axis inertial momentum grows linearly, as expected. However, an oscillation is seen in the x - and y -axes inertial momentum. This cannot happen physically and is an artifact of numerical integration. This can be confirmed by running the same simulation with a time step of 1 s, as shown in the lower plot.

The oscillation magnitude in the lower plot is much larger than in the first plot. Whenever you do this kind of simulation it is always important to verify that angular momentum is conserved. When an integrator that does not have error correction is used, such as RK4, you should change the time step and observe the inertial angular momentum. If you use an error-correcting integrator, such as RK45 or MATLAB®'s ode113, you can change the error tolerance and observe the result. If tightening the error tolerance does not improve the result, there is an error in the dynamics.

6.5. Gyrostat

A gyrostat is a spacecraft that is rigid except for symmetric rotors that spin about axes fixed to the body frame. Many spacecraft are gyrostats since reaction wheels can control the spacecraft's attitude without the need to expend propellant. A gyrostat is used to represent spacecraft with reaction wheels. The momentum for a gyrostat is

$$H = A \left(I\omega + \sum_k u_k J_k (\Omega_k + u_k^T \omega) \right) \quad (6.21)$$

where Ω_k is the angular rate of the k th wheel about its spin axis, ω are the body rates, J_k is the spin-axis moment of inertia, u_k is the unit vector of the wheel spin axis measured in the body frame, and I is the inertia of the rigid body excluding the wheels. The equations of motion are then

$$T = I\dot{\omega} + \omega^\times \left(I\omega + \sum_k u_k J_k (\Omega_k + u_k^T \omega) \right) + \sum_k u_k J_k (\dot{\Omega}_k + u_k^T \dot{\omega}) \quad (6.22)$$

$$T_k = J_k (\dot{\Omega}_k + u_k^T \dot{\omega}) \quad (6.23)$$

The transverse axes' inertias are lumped into the core body inertia. T is the body-fixed torque on the spacecraft and T_k is the scalar torque on each wheel. These equations give $3 + n$ equations of motion. To integrate these equations you need first to solve for the angular acceleration of the core, and then substitute the angular-acceleration vector into the reaction-wheel equations.

$$\dot{\omega} = I^{-1} \left(T - \omega^\times \left(I\omega + \sum_k u_k J_k (\Omega_k + u_k^T \omega) \right) - \sum_k u_k T_k \right) \quad (6.24)$$

$$\dot{\Omega}_k = \frac{T_k}{J_k} - u_k^T \dot{\omega} \quad (6.25)$$

Spacecraft with reaction and momentum wheels are the most popular type of spacecraft design today. Some examples are

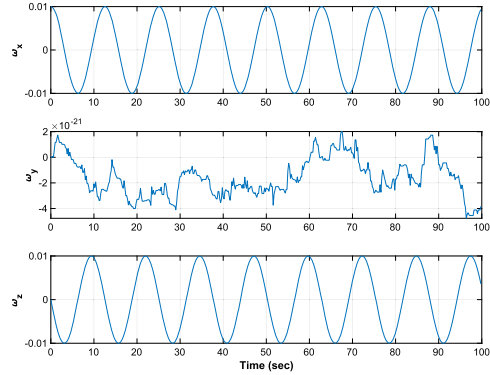
- Boeing HS702 communications Satellite;
- Lockheed Martin A2100 communications Satellite;
- Lockheed Martin GPS IIR navigation satellite;
- NASA/JPL Cassini scientific satellite.

Example 6.3 shows the effect of having a bias momentum on a spacecraft. The spacecraft has a 1-N m² wheel aligned with the y -axis spinning at 100 rad/s. One of the transverse rates (about the x -axis) is set to 0.01 rad/s. The spacecraft has a diagonal inertia matrix that eliminates all Euler coupling between the axes. The momentum bias introduces coupling that causes the angular momentum to be transferred between the x - and z -axes thus maintaining the overall pointing orientation of the spacecraft. The oscillation is called *nutation*. The y -axis rate in the plot is due to a numerical integration error.

```

1 dRHS = RHSGyrost;
2 dRHS.uWheel = [0;1;0];
3 dRHS.inrWheel = 1;
4 dRHS.inr = 200*eye(3);
5 dRHS.torqueWheel = 0;
6 dRHS.friction = struct('coulomb',0,'damping',
    ,0,'kCoulomb',0);
7 tEnd = 100;
8 dT = 0.2;
9 nSim = floor(tEnd/dT);
10 xP = zeros(3,nSim);
11 x = [[1;0;0;0];[0.01;0;0];100];
12 for k = 1:nSim
13     xP(:,k) = x(5:7);
14     x = RK4(@RHSGyrost,x,dT,0,dRHS);
15 end
16 yL = {'\omega_x' '\omega_y' '\omega_z'};
17 TimeHistory(0:(nSim-1)*dT,xP,yL,'Momentum_Bias')

```



Example 6.3: Momentum-bias spacecraft nutation.

6.6. Dual spin

A special case of a gyrost is a dual-spin spacecraft. This has one spun part and one part rotating at orbit rate. The dynamical equations are

$$T = I\dot{\omega} + \omega^\times (I\omega + uJ(\Omega + u^T\omega)) + uJ(\dot{\Omega} + u^T\dot{\omega}) \quad (6.26)$$

$$T_w = J(\dot{\Omega} + u^T\dot{\omega}) \quad (6.27)$$

where

$$u = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \quad (6.28)$$

Assume that the body rates are

$$\omega = \begin{bmatrix} \omega_x \\ \omega_y + \omega_o \\ \omega_z \end{bmatrix} \quad (6.29)$$

where the body rates are small, ω_o is the orbit rate, and γ is along the orbit normal. Assume that the inertia matrix is diagonal and that Ω is constant. We then get

$$T = (I + uJu^T)\dot{\omega} + \omega^\times (I\omega + uJ(\Omega + u^T\omega)) \quad (6.30)$$

The wheel torque compensates for $\dot{\omega}_\gamma$. Note that

$$uJu^T = \begin{bmatrix} 0 & 0 & 0 \\ 0 & J & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad (6.31)$$

The linearized equations are

$$T_x = I_x\dot{\omega}_x - \omega_z(\omega_o(I_y - I_z) + \Omega J) \quad (6.32)$$

$$T_\gamma = I_\gamma\dot{\omega}_\gamma \quad (6.33)$$

$$T_z = I_z\dot{\omega}_z + \omega_x(\omega_o(I_y - I_x) + \Omega J) \quad (6.34)$$

where I is now the total spacecraft inertia, including the wheel.

$$\begin{bmatrix} \omega_x \\ \omega_z \end{bmatrix} = \frac{\begin{bmatrix} s & \omega_o(I_y - I_z) + \Omega J \\ -\omega_o(I_y - I_x) - \Omega J & s \end{bmatrix}}{s^2 + (\omega_o(I_y - I_z) + \Omega J)(\omega_o(I_y - I_x) + \Omega J)} \begin{bmatrix} a_x \\ a_z \end{bmatrix} \quad (6.35)$$

The nutational modes are pure oscillatory as long as these two conditions are met

$$\Omega J > \omega_o(I_y - I_x) \quad (6.36)$$

$$\Omega J > \omega_o(I_y - I_z) \quad (6.37)$$

6.7. Gravity gradient

An Earth-pointing spacecraft, with the y -axis along the negative orbit normal (the LVLH frame), is modeled by the following dynamical equation

$$I\dot{\omega} + \omega^\times I\omega = T_{gg} + T \quad (6.38)$$

The coordinate frame, which is rotating at a steady rotation rate n about the $-y$ -axis is shown in Fig. 6.2.

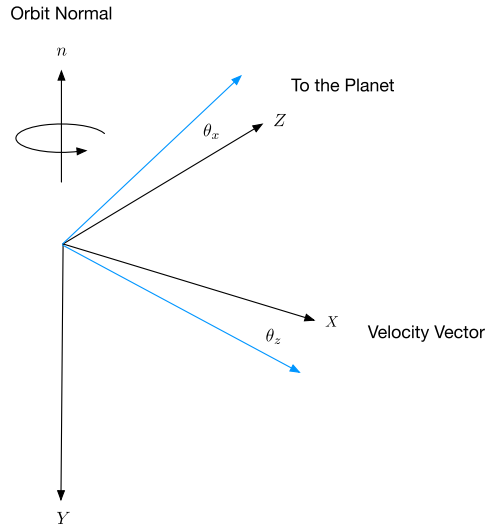


Figure 6.2 Rotating coordinate frame.

This leads to the following kinematical equations by inspection

$$\omega = \begin{bmatrix} \dot{\theta}_x - n\theta_z \\ \dot{\theta}_y - n \\ \dot{\theta}_z + n\theta_x \end{bmatrix} \quad (6.39)$$

and

$$\dot{\omega} = \begin{bmatrix} \ddot{\theta}_x - n\dot{\theta}_z \\ \ddot{\theta}_y \\ \ddot{\theta}_z + n\dot{\theta}_x \end{bmatrix} \quad (6.40)$$

where n is the orbit rate and where the angles are all small angles from the rotating frame. We will first do the traditional derivation, and then do one that includes the off-diagonal inertia terms.

The gravity-gradient disturbance torque arises from the inertia matrix. The vector torque is

$$T = 3\mu \frac{r^\times I r}{|r|^5} \quad (6.41)$$

where r is the vector to the satellite from the center of the Earth, r is along the $-z$ -axis of the rotating frame. As long as μ and r are in consistent units, the distance units that are used drop out and the result is $1/s^2$. The torque will then be in the units determined by the units of the inertia matrix. Substituting, we get

$$T = 3n^2 u^\times I u \quad (6.42)$$

where the unit vector

$$u = \frac{r}{|r|} \quad (6.43)$$

is approximately

$$u = \begin{bmatrix} -\theta_y \\ \theta_x \\ 1 \end{bmatrix} \quad (6.44)$$

The linearized gravity-gradient torque is

$$T_{gg} = 3n^2 \begin{bmatrix} (I_z - I_y)\theta_x \\ (I_z - I_x)\theta_y \\ 0 \end{bmatrix} \quad (6.45)$$

and n is the orbit rate. Assume that the inertia matrix is diagonal and use the linearized body rates. Euler's equations become

$$T_x = I_x \ddot{\theta}_x + n \dot{\theta}_z (I_y - I_z - I_x) - n^2 \theta_x (I_z - I_y) \quad (6.46)$$

$$T_y = I_y \ddot{\theta}_y \quad (6.47)$$

$$T_z = I_z \ddot{\theta}_z + n \dot{\theta}_x (I_z + I_x - I_y) + n^2 \theta_z (I_y - I_x) \quad (6.48)$$

Adding the gravity-gradient torque we get three second-order differential equations

$$T_x = I_x \ddot{\theta}_x + n \dot{\theta}_z (I_y - I_z - I_x) - 4n^2 \theta_x (I_z - I_y) \quad (6.49)$$

$$T_y = I_y \ddot{\theta}_y + 3n^2 (I_x - I_z) \theta_y \quad (6.50)$$

$$T_z = I_z \ddot{\theta}_z + n \dot{\theta}_x (I_z + I_x - I_y) + n^2 \theta_z (I_y - I_x) \quad (6.51)$$

All of the stiffness terms, that is terms that include angle, must have positive coefficients for stability. The x and z equations are coupled. The y equation is a harmonic oscillator.

Let

$$k_x = 4n^2 \frac{I_y - I_z}{I_x} \quad (6.52)$$

$$k_y = 3n^2 \frac{I_x - I_z}{I_y} \quad (6.53)$$

$$k_z = n^2 \frac{I_y - I_x}{I_z} \quad (6.54)$$

$$c_x = n \frac{I_y - I_z - I_x}{I_x} \quad (6.55)$$

$$c_z = n \frac{I_z + I_x - I_y}{I_z} \quad (6.56)$$

Since y is decoupled, $I_x > I_z$ is required. The frequency of oscillation for y is

$$\Omega_y = n\sqrt{3k_y} \quad (6.57)$$

The xz frequencies are the libration frequencies and are found in the following equations

$$a_x = \ddot{\theta}_x + c_x \dot{\theta}_z + k_x \theta_x \quad (6.58)$$

$$a_z = \ddot{\theta}_z + c_z \dot{\theta}_x + k_z \theta_z \quad (6.59)$$

The full 3-axis system in state-space form is

$$\begin{bmatrix} \dot{\theta}_x \\ \dot{\theta}_y \\ \dot{\theta}_z \\ \ddot{\theta}_x \\ \ddot{\theta}_y \\ \ddot{\theta}_z \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ -k_x & 0 & 0 & 0 & 0 & -c_x \\ 0 & -k_y & 0 & 0 & 0 & 0 \\ 0 & 0 & -k_z & -c_z & 0 & 0 \end{bmatrix} \begin{bmatrix} \theta_x \\ \theta_y \\ \theta_z \\ \dot{\theta}_x \\ \dot{\theta}_y \\ \dot{\theta}_z \end{bmatrix} \quad (6.60)$$

Write the equations for x and z in the s domain

$$\begin{bmatrix} s^2 + k_x & c_x s \\ c_z s & s^2 + k_z \end{bmatrix} \begin{bmatrix} \theta_x \\ \theta_z \end{bmatrix} = \begin{bmatrix} a_x \\ a_z \end{bmatrix} \quad (6.61)$$

where a_x and a_z are the angular accelerations in x and z . The denominator will be quadratic in s^2 and the analytical solution is straightforward.

$$\begin{bmatrix} \theta_x \\ \theta_z \end{bmatrix} = \frac{\begin{bmatrix} s^2 + k_z & -c_x s \\ -c_z s & s^2 + k_x \end{bmatrix}}{s^4 + (k_x + k_z - c_x c_z) s^2 + k_x k_z} \begin{bmatrix} a_x \\ a_z \end{bmatrix} \quad (6.62)$$

Write the denominator as

$$s^4 + b_1 s^2 + b_0 \quad (6.63)$$

The eigenvalues are

$$s^2 = -\frac{b_1}{2} \pm \sqrt{\frac{b_1^2}{4} - b_0} \quad (6.64)$$

For s to have pure imaginary roots, the roots for s^2 must be negative. This requires

$$b_0 > 0 \quad (6.65)$$

$$b_1 > 0 \quad (6.66)$$

$$b_1 > 2b_0 \quad (6.67)$$

The last condition insures that the term in the square root is positive. Of course, if this condition is true then the second is also true. If these criteria are met, the system is an undamped oscillator. The first requires that

$$I_x > I_y > I_z \quad (6.68)$$

To provide damping we need to add

$$a_x = -2\zeta \sqrt{k_x} \dot{\theta}_x \quad (6.69)$$

Generally, the problem is we have bus inertia, I_0 to which we want to add a boom to give us gravity-gradient stability. Define

$$\dot{\omega} = \ddot{\theta} + D\dot{\theta} \quad (6.70)$$

$$\omega = \dot{\theta} + D\theta + N \quad (6.71)$$

where

$$D = \begin{bmatrix} 0 & 0 & -n \\ 0 & 0 & 0 \\ n & 0 & 0 \end{bmatrix} \quad (6.72)$$

$$N = \begin{bmatrix} 0 \\ -n \\ 0 \end{bmatrix} \quad (6.73)$$

The linearized Euler's equation with gravity gradient becomes

$$T = I\ddot{\theta} + ID\dot{\theta} + (\dot{\theta} + D\theta + N)^\times I(\dot{\theta} + D\theta + N) - G\theta \quad (6.74)$$

Removing products of small quantities

$$T = I\ddot{\theta} + ID\dot{\theta} + \dot{\theta}^\times IN + (D\theta)^\times IN + N^\times I\dot{\theta} + N^\times ID\theta - G\theta \quad (6.75)$$

We use the matrix identities

$$(D\theta)^\times = D\theta^\times D^T \quad (6.76)$$

$$a^\times b = -b^\times a \quad (6.77)$$

and

$$G = 3n^2 \begin{bmatrix} I_z - I_y & I_{xy} & 0 \\ I_{xy} & I_z - I_x & 0 \\ -I_{xz} & I_{yz} & 0 \end{bmatrix} \quad (6.78)$$

results in

$$T = I\ddot{\theta} + ID\dot{\theta} - (IN)^\times \dot{\theta} - (D^T IN)^\times D\theta + N^\times I\dot{\theta} + N^\times ID\theta - G\theta \quad (6.79)$$

We can then put this into state-space form to compute the eigenvalues.

$$\begin{bmatrix} \dot{\theta} \\ \ddot{\theta} \end{bmatrix} = \begin{bmatrix} 0 & E \\ -I^{-1}(N^\times ID - G - (IN)^\times D) & -I^{-1}(ID - (IN)^\times + N^\times I) \end{bmatrix} \begin{bmatrix} \theta \\ \dot{\theta} \end{bmatrix} \quad (6.80)$$

where E is a 3-by-3 identity matrix. Example 6.4 compares the two formulations.

Higher-order models for gravity gradient are also available [2]. The following includes only the J_2 term.

$$T_g = \frac{3\mu}{|r|^3} u_r^\times I u_r \quad (6.81)$$

$$+ \frac{\mu J_2 R_p^2}{2|r|^5} [30u_r^T u_n (u_r^\times I u_n + u_n^\times I u_r) + (15 - 105(u_r^T u_n)^2) u_n^\times I u_r + 6u_n^\times I u_n]$$

where J_2 is the gravitational harmonic, u_r is the unit vector from the spacecraft center-of-mass to the center of the planet, R_p is the planet radius, and u_n is the unit vector along the rotation axis of the planet. Since J_2 is small, this term is very small, particularly when $|r| \gg R_p$.

6.8. Nutation dynamics

A spinning spacecraft is modeled by Euler's equation

$$I\dot{\omega} + \omega^\times I\omega = T \quad (6.82)$$

```

1 %% Gravity Gradient
2
3 inr = [3 0 0;0 2 0;0 0 1];
4
5 n = 2*pi/(90*60);
6
7 nSq = n^2;
8
9 kX = 4*nSq*(inr(2,2) - inr(3,3))/inr(1,1);
10 kY = 3*nSq*(inr(1,1) - inr(3,3))/inr(2,2);
11 kZ = nSq*(inr(2,2) - inr(1,1))/inr(3,3);
12 cX = n*(inr(2,2) - inr(3,3) - inr(1,1))/inr(1,1);
13 cZ = n*(inr(3,3) + inr(1,1) - inr(2,2))/inr(3,3);
14
15 a = [zeros(3,3) eye(3);-diag([kX kY kZ]) ...
16     [0 0 -cX;0 0 0;-cZ 0 0]];
17
18 disp('Eigenvalues for the matrix')
19 eig(a)
20
21 disp('Roots for roll/yaw')
22 roots([1 0 (kX+kZ - cX*cZ) 0 kX*kZ])
23
24 theta = [0.001;0.002;0.003];
25 thetaDot = theta/100;
26
27 N = [0;-n;0];
28 D = [0 0 -n;0 0 0;n 0 0];
29 G = 3*n^2*[inr(3,3)-inr(2,2) inr(1,2) 0;...
30 inr(1,2) inr(3,3)-inr(1,1) 0;...
31 -inr(1,3) inr(2,3) 0];
32
33 a = [zeros(3,3) eye(3);-inr\((Skew(N)*inr*D - G -
34     Skew(inr*N)*D) ...
35 -inr\((inr*D - Skew(inr*N) + Skew(N)*inr))];
36
37 disp('Eigenvalues for the full matrix')
38 eig(a)

```

```

1 Eigenvalues for the matrix
2
3 ans =
4
5 -8.9425e-04 + 0.0000e+00i
6 8.9425e-04 + 0.0000e+00i
7 5.4210e-20 + 1.7482e-03i
8 5.4210e-20 - 1.7482e-03i
9 0.0000e+00 + 2.0153e-03i
10 0.0000e+00 - 2.0153e-03i
11
12 Roots for roll/yaw
13
14 ans =
15
16 3.3881e-19 + 1.7482e-03i
17 3.3881e-19 - 1.7482e-03i
18 -8.9425e-04 + 0.0000e+00i
19 8.9425e-04 + 0.0000e+00i
20
21 Eigenvalues for the full matrix
22
23 ans =
24
25 8.9425e-04 + 0.0000e+00i
26 -8.9425e-04 + 0.0000e+00i
27 -2.8309e-101 + 1.7482e-03i
28 -2.8309e-101 - 1.7482e-03i
29 0.0000e+00 + 2.0153e-03i
30 0.0000e+00 - 2.0153e-03i

```

Example 6.4: Two formulations for gravity gradient. The second can accommodate off-diagonal terms.

where T is torque, I is inertia, and ω is the body angular rate expressed in the body frame. Assume that the spacecraft inertia matrix is diagonal. The equations become

$$I_x \dot{\omega}_x + \omega_y \omega_z (I_z - I_y) = T_x \quad (6.83)$$

$$I_y \dot{\omega}_y + \omega_x \omega_z (I_x - I_z) = T_y \quad (6.84)$$

$$I_z \dot{\omega}_z - \omega_x \omega_y (I_x - I_y) = T_z \quad (6.85)$$

Define

$$k_x = \frac{I_z - I_y}{I_x} \quad (6.86)$$

$$k_y = \frac{I_x - I_z}{I_y} \quad (6.87)$$

$$k_z = \frac{I_x - I_y}{I_z} \quad (6.88)$$

and

$$a_x = \frac{T_x}{I_x} \quad (6.89)$$

$$a_y = \frac{T_y}{I_y} \quad (6.90)$$

$$a_z = \frac{T_z}{I_z} \quad (6.91)$$

we then get

$$\dot{\omega}_x + \omega_y \omega_z k_x = a_x \quad (6.92)$$

$$\dot{\omega}_y + \omega_x \omega_z k_y = a_y \quad (6.93)$$

$$\dot{\omega}_z - \omega_x \omega_y k_z = a_z \quad (6.94)$$

Assume also that the angular-momentum vector is the sum of a set of small angular rates and a large spin rate about the y -axis.

$$\omega = \begin{bmatrix} \omega_x \\ \omega_y \\ \omega_z \end{bmatrix} + \begin{bmatrix} 0 \\ \Omega \\ 0 \end{bmatrix} \quad (6.95)$$

where ω is now a vector of small angular rates and Ω is constant. We then get linearized dynamical equations

$$\dot{\omega}_x + k_x \Omega \omega_z = a_x \quad (6.96)$$

$$\dot{\omega}_y = a_y \quad (6.97)$$

$$\dot{\omega}_z - k_z \Omega \omega_x = a_z \quad (6.98)$$

where $k = k_x \Omega$. The nutational equations are

$$\begin{bmatrix} s & k_x \Omega \\ -k_z \Omega & s \end{bmatrix} \begin{bmatrix} \omega_x \\ \omega_z \end{bmatrix} = \begin{bmatrix} a_x \\ a_z \end{bmatrix} \quad (6.99)$$

solving

$$\begin{bmatrix} \omega_x \\ \omega_z \end{bmatrix} = \frac{\begin{bmatrix} s & -k_x \Omega \\ k_z \Omega & s \end{bmatrix}}{s^2 + k_x k_z \Omega^2} \begin{bmatrix} a_x \\ a_z \end{bmatrix} \quad (6.100)$$

the nutation frequency is

$$\omega_n = \Omega \sqrt{k_x k_z} \quad (6.101)$$

A major-axis spinner is

$$I_y > I_x > I_z \quad (6.102)$$

A minor-axis spinner

$$I_x > I_z > I_y \quad (6.103)$$

An intermediate-axis spinner is, for example,

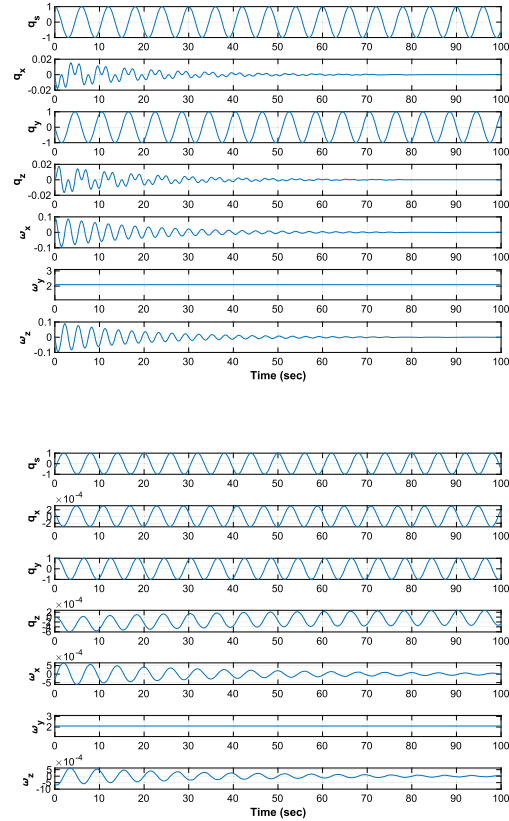
$$I_x > I_y > I_z \quad (6.104)$$

For a major- or minor-axis spinner, $k_x k_z > 0$, and the spacecraft spin is stable. For an intermediate-axis spinner, $k_x k_z < 0$, and therefore it is unstable. The two cases are shown in Example 6.5. The attitude motion does not damp with the minor-axis spinner.

```

1 %% Nutation
2
3 clear d
4
5 % MajorAxis
6 d.inr = diag([1 2 1]); % Major axis
7 d.damp = 0.1;
8 x = [1;0;0;0;0.1;20*pi/30;0];
9 n = 1000;
10 xP = zeros(7,n);
11 dT = 0.1;
12 for k = 1:n
13     xP(:,k) = x;
14     x = RK4(@RHS,x,dT,0,d);
15 end
16
17 t = (0:n-1)*dT;
18 yL = {'q_s' 'q_x' 'q_y' 'q_z' ...
19       '\omega_x' '\omega_y' '\omega_z' ...};
20 TimeHistory(t,xP,yL,'MajorAxis');
21
22 % Minor axis
23 d.inr = diag([2 1 2]); % Major axis
24
25 for k = 1:n
26     xP(:,k) = x;
27     x = RK4(@RHS,x,dT,0,d);
28 end
29
30 TimeHistory(t,xP,yL,'MinorAxis');
31
32 function xDot = RHS(x,~,d)
33
34 q = x(1:4);
35 omega = x(5:7);
36 t = -d.damp*x(5);
37 t = [x(5);0;0];
38
39 omegaDot = d.inr \ (t - Cross(omega,d.inr*omega));
40
41 xDot = [QToBDot(q,omega);omegaDot];
42
43 end

```



Example 6.5: Major- and minor-axis spin.

6.9. Planar slosh model

Fig. 6.3 shows the planar slosh model. It consists of a slosh mass and a central body with inertia and mass.

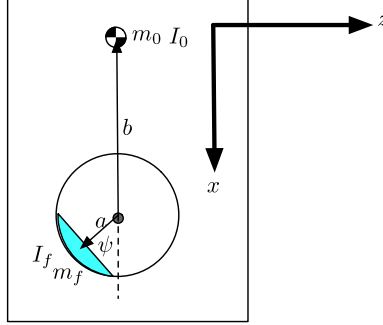


Figure 6.3 The planar slosh system.

The equations of motion are derived using the Lagrangian, which is the difference between the kinetic and potential energies.

$$L = T - V \quad (6.105)$$

The equations of motion are found from

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) - \frac{\partial L}{\partial q_i} = 0 \quad (6.106)$$

where q is an independent coordinate.

The position of the center-of-mass with respect to the fuel tank center-of-mass is

$$\vec{r} = (x - b)\hat{i} + z\hat{k} \quad (6.107)$$

The velocity of the point is

$$\vec{v} = \dot{\vec{r}} - \omega \times \vec{r} \quad (6.108)$$

where

$$\omega^\times = \begin{bmatrix} 0 & 0 & -\dot{\theta} \\ 0 & 0 & 0 \\ \dot{\theta} & 0 & 0 \end{bmatrix} \quad (6.109)$$

which is

$$\vec{v} = (\dot{x} + \dot{\theta}z)\hat{i} + (\dot{z} + \dot{\theta}(b - x))\hat{k} \quad (6.110)$$

With $v_x = \dot{x} + \dot{\theta}z$ and $v_z = \dot{z} - x\dot{\theta}$ we can write this as

$$\vec{v} = v_x \hat{i} + (v_z + b\dot{\theta}) \hat{k} \quad (6.111)$$

The position of the slosh mass is

$$\vec{r}_f = (x - a \cos \psi) \hat{i} + (z + a \sin \psi) \hat{k} \quad (6.112)$$

so

$$\vec{v}_f = \dot{\vec{r}}_f - \omega^\times \vec{r}_f \quad (6.113)$$

or

$$\vec{v}_f = (\dot{x} + a\dot{\psi} \sin \psi) \hat{i} + (\dot{z} + a\dot{\psi} \cos \psi) \hat{k} + \dot{\theta}(a \cos \psi - x) \hat{k} + \dot{\theta}(z + a \sin \psi) \hat{i} \quad (6.114)$$

Rearranging

$$\vec{v}_f = (\dot{x} + a\dot{\psi} \sin \psi + \dot{\theta}(z + a \sin \psi)) \hat{i} + (\dot{z} + a\dot{\psi} \cos \psi + \dot{\theta}(a \cos \psi - x)) \hat{k} \quad (6.115)$$

or

$$\vec{v}_f = (\dot{x} + a(\dot{\theta} + \dot{\psi}) \sin \psi + z\dot{\theta}) \hat{i} + (\dot{z} + a(\dot{\psi} + \dot{\theta}) \cos \psi - x\dot{\theta}) \hat{k} \quad (6.116)$$

or

$$\vec{v}_f = (v_x + a(\dot{\theta} + \dot{\psi}) \sin \psi) \hat{i} + (v_z + a(\dot{\psi} + \dot{\theta}) \cos \psi) \hat{k} \quad (6.117)$$

The kinetic energy is

$$T = \frac{1}{2} m_0 \vec{r}^2 + \frac{1}{2} m_f \vec{r}_f^2 + \frac{1}{2} I_0 \dot{\theta}^2 + \frac{1}{2} I_f \dot{\theta}_f^2 \quad (6.118)$$

or

$$\begin{aligned} T = & \frac{1}{2} m_0 (v_x^2 + (v_z + b\dot{\theta})^2) \\ & + \frac{1}{2} m_f ((v_x + a(\dot{\theta} + \dot{\psi}) \sin \psi)^2 + (v_z + a(\dot{\psi} + \dot{\theta}) \cos \psi)^2) \\ & + \frac{1}{2} I_0 \dot{\theta}^2 + \frac{1}{2} I_f (\dot{\theta} + \dot{\psi})^2 \end{aligned} \quad (6.119)$$

Expanding the squares

$$\begin{aligned} T = & \frac{1}{2} m_0 (v_x^2 + v_z^2 + 2v_z b\dot{\theta} + b^2 \dot{\theta}^2) \\ & + \frac{1}{2} m_f (v_z^2 + 2v_z a(\dot{\psi} + \dot{\theta}) \cos \psi) \end{aligned} \quad (6.120)$$

$$\begin{aligned}
& + \frac{1}{2}m_f(v_x^2 + 2v_x a(\dot{\theta} + \dot{\psi}) \sin \psi) \\
& + \frac{1}{2}m_f a^2(\dot{\theta}^2 + 2\dot{\psi}\dot{\theta} + \dot{\psi}^2) \\
& + \frac{1}{2}I_0\dot{\theta}^2 + \frac{1}{2}I_f(\dot{\theta}^2 + 2\dot{\theta}\dot{\psi} + \dot{\psi}^2)
\end{aligned}$$

The potential energy is due to the spring at the slosh hinge and is

$$V = \frac{1}{2}k\psi^2 \quad (6.121)$$

and the Raleigh dissipation function is

$$R = \frac{1}{2}c\dot{\psi}^2 \quad (6.122)$$

The equations are

$$0 = (m_0 + m_f)\dot{v}_x + m_f a(\ddot{\theta} + \ddot{\psi}) \sin \psi + m_f a(\dot{\theta} + \dot{\psi})\dot{\psi} \cos \psi \quad (6.123)$$

$$0 = (m_0 + m_f)\dot{v}_z + m_f a(\ddot{\theta} + \ddot{\psi}) \cos \psi - m_f a(\dot{\theta} + \dot{\psi})\dot{\psi} \sin \psi + m_0 b\ddot{\theta} \quad (6.124)$$

$$0 = (I_0 + I_f + m_f a^2)\ddot{\theta} + m_f a(\dot{v}_x \sin \psi + \dot{v}_z \cos \psi) \quad (6.125)$$

$$\begin{aligned}
& + m_f a\dot{\psi}(v_x \cos \psi - v_z \sin \psi) + (I_f + m_f a^2)\ddot{\psi} \\
0 = & (I_f + m_f a^2)(\ddot{\psi} + \ddot{\theta}) + m_f a(v_x \cos \psi - v_z \sin \psi)\dot{\psi} \\
& - a(\dot{\theta} + \dot{\psi})(v_z \sin \psi - v_x \cos \psi)
\end{aligned} \quad (6.126)$$

To the last equation, we add the damping and spring stiffness

$$M_\psi = -c\dot{\psi} - k\psi \quad (6.127)$$

Linearized equations

$$0 = (m_0 + m_f)\dot{v}_x \quad (6.128)$$

$$0 = (m_0 + m_f)\dot{v}_z + m_f a(\ddot{\theta} + \ddot{\psi}) + m_0 b\ddot{\theta} \quad (6.129)$$

$$0 = (I_0 + I_f + m_f a^2)\ddot{\theta} + m_f a\dot{v}_z + (I_f + m_f a^2)\ddot{\psi} \quad (6.130)$$

$$0 = (I_f + m_f a^2)(\ddot{\psi} + \ddot{\theta}) + c\dot{\psi} + k\psi \quad (6.131)$$

An alternative is to use quasicordinates [3,4]. Lagrange's equation for the spacecraft angle and velocities are found from

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{v}} \right) + \omega^\times \frac{\partial L}{\partial \dot{v}} = 0 \quad (6.132)$$

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \vec{\omega}} \right) + \vec{\omega}^\times \frac{\partial L}{\partial \vec{\omega}} + \vec{v}^\times \frac{\partial L}{\partial \vec{v}} = 0 \quad (6.133)$$

where

$$\vec{a}^\times = \begin{bmatrix} 0 & -a_z & a_y \\ a_z & 0 & -a_x \\ -a_y & a_x & 0 \end{bmatrix} \quad (6.134)$$

Neither equation has the partial with respect to the generalized coordinate. The dynamical equations are

$$F_x(m_0 + m_f)\dot{v}_x + m_f a(\ddot{\theta} + \ddot{\psi}) \sin \psi \quad (6.135)$$

$$+ m_f a(\dot{\theta} + \dot{\psi})\dot{\psi} \cos \psi + m_f a(\dot{\theta} + \dot{\psi})^2 \cos \psi$$

$$F_z = (m_0 + m_f)(\dot{v}_z - \dot{\theta}v_x) + m_f a(\ddot{\theta} + \ddot{\psi}) \cos \psi \quad (6.136)$$

$$- m_f a(\dot{\theta} + \dot{\psi})\dot{\psi} \sin \psi$$

$$+ m_0 b\ddot{\theta} - m_f a(\dot{\theta} + \dot{\psi})^2 \cos \psi$$

$$M + bF_z = (I_0 + m_0 b^2)\ddot{\theta} + mb(\dot{v}_z - \dot{\theta}v_x) - c\dot{\psi} \quad (6.137)$$

$$0 = (I_f + m_f a^2)(\ddot{\psi} + \ddot{\theta}) + m_f a(v_x \cos \psi - v_z \sin \psi)\dot{\psi} \quad (6.138)$$

$$- (\dot{\theta} + \dot{\psi})(v_z \sin \psi - v_x \cos \psi) + c\dot{\psi} + k\psi$$

6.10. N-body hub with single degree-of-freedom hinges

The N-body hub model can be specialized to a situation in which each hinge has only one degree-of-freedom. An example with two appendages is shown in Fig. 6.4.

The axes of rotation for the two appendages (the momentum wheel and telescope) are u_1 and u_2 . The corresponding angular rate vectors are

$$\omega_0 = \omega_0 \quad (6.139)$$

$$\omega_k = B_k^T \omega_0 + \dot{\theta}_k u_k \quad (6.140)$$

where θ_k is the angle about the unit vector u_k , u_k is fixed in the body 0 frame, B_k transforms from the k frame to the 0 frame, and B_0 is the identity matrix. The corresponding accelerations are

$$\dot{\omega}_0 = \dot{\omega}_0 \quad (6.141)$$

$$\dot{\omega}_k = \dot{B}_k^T \omega_0 + B_k^T \dot{\omega}_0 + \ddot{\theta}_k u_k \quad (6.142)$$

The equations of the core attitude are

$$\dot{q} = f(\omega_0, q) \quad (6.143)$$

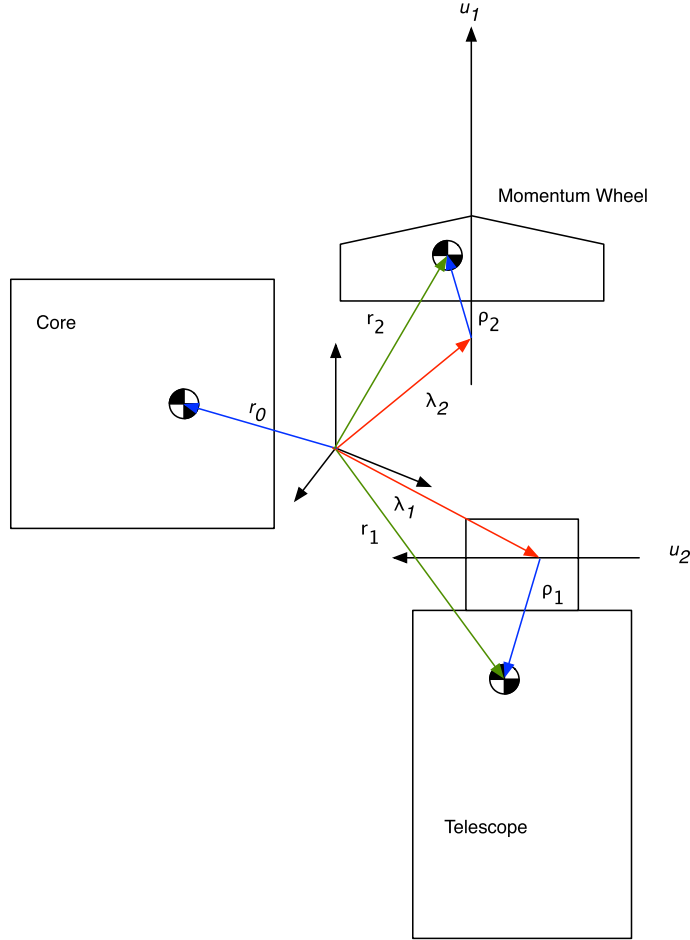


Figure 6.4 Dynamical model with two one-degree-of-freedom hinges for the momentum wheel and the telescope. Unlike a gyrostat, these are not necessarily rotating about their center-of-mass.

where q is the attitude quaternion from the ECI frame to the core body frame. Let B_k transform from the body k frame to the body 0 frame. Define

$$r_k = \lambda_k + B_k \rho_k \quad (6.144)$$

where A is the matrix that transforms from the 0 frame to the inertial frame and A^T corresponds to q .

$$H = A \sum_k B_k I_k \omega_k + \sum_k m_k D_k^\times \dot{D}_k \quad (6.145)$$

where I_k is the inertia matrix about the k body center-of-mass written in the k frame. This is conserved in the absence of external torques. Expanding the second term

$$H = A \sum_k \left(B_k I_k \omega_k + m_k d_k^\times (\omega_0^\times d_k + \dot{d}_k) \right) \quad (6.146)$$

Let

$$H = Ah \quad (6.147)$$

Then,

$$T = \omega_0^\times h + \dot{h} \quad (6.148)$$

Expanding

$$T = \omega_0^\times h + \sum_k \left(\dot{B}_k I_k \omega_k + B_k I_k \dot{\omega}_k + m_k (\dot{d}_k^\times \omega_0^\times d_k + d_k^\times \omega_0^\times \dot{d}_k - d_k^\times d_k^\times \dot{\omega}_0 + d_k^\times \ddot{d}_k) \right) \quad (6.149)$$

The angular-acceleration term is

$$\begin{aligned} \dot{\omega}_k &= \dot{B}_k^T \omega_0 + B_k^T \dot{\omega}_0 + \ddot{\theta}_k u_k & k > 0 \\ \dot{\omega}_k &= \dot{\omega}_0 & k = 0 \end{aligned} \quad (6.150)$$

where T is the torque on the entire spacecraft expressed in the 0 frame. The appendage equations are found by computing the reaction to the torques at the hinge points. The total angular momentum at the hinge is

$$H_k = AB_k I_k \omega_k + m_k (AB_k \rho_k)^\times \dot{D}_k \quad (6.151)$$

Expanding

$$H_k = Ah_k \quad (6.152)$$

$$h_k = B_k I_k \omega_k + m_k (B_k \rho_k)^\times (\omega_0^\times d_k + \dot{d}_k) \quad (6.153)$$

The scalar torque equation about u_k is

$$\begin{aligned} T_k &= u_k^T [\omega_0^\times h_k \\ &\quad + \dot{B}_k I_k \omega_k + B_k I_k \dot{\omega}_k + m_k (\dot{B}_k \rho_k)^\times (\omega_0^\times d_k + \dot{d}_k) \\ &\quad + m_k (B_k \rho_k)^\times (\omega_0^\times \dot{d}_k - d_k^\times \dot{\omega}_0 + \ddot{d}_k)] \end{aligned} \quad (6.154)$$

Expanding d_k and its derivatives

$$d_k = \lambda_k + B_k \rho_k - c \quad (6.155)$$

$$d_k = \sum_j s_{kj}(\lambda_j + B_j \rho_j) \quad (6.156)$$

$$\dot{d}_k = \sum_j s_{kj} B_j \Omega_j^\times \rho_j \quad (6.157)$$

$$\ddot{d}_k = \sum_j s_{kj} B_j (\Omega_j^\times \Omega_j^\times \rho_j - \rho_j^\times \dot{\Omega}_j) \quad (6.158)$$

$$s = \begin{bmatrix} 1 - \mu_0 & -\mu_1 & -\mu_2 \\ -\mu_0 & 1 - \mu_1 & -\mu_2 \\ -\mu_0 & -\mu_1 & 1 - \mu_2 \end{bmatrix} \quad (6.159)$$

The final equation will be

$$\begin{bmatrix} A_{00} & A_{01} & A_{02} \\ A_{10} & a_{11} & a_{12} \\ A_{20} & a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} \dot{\omega}_0 \\ \ddot{\theta}_1 \\ \ddot{\theta}_2 \end{bmatrix} = \begin{bmatrix} E_0 \\ e_1 \\ e_2 \end{bmatrix} \quad (6.160)$$

where A_{00} is a 3×3 matrix and is the inertia matrix for the entire vehicle, A_{01} and A_{02} are 3×1 matrices, E_0 is 3×1 , and e_1 and e_2 are scalars. The inertia matrix–vector and matrix terms are

$$A_{00} = I_0 + \sum_{k>0} (B_k I_k B_k^T - m_k d_k^\times d_k^\times) \quad (6.161)$$

The remaining terms in the inertia matrix are

$$\sum_k \left(B_k I_k u_k \ddot{\theta}_k - m_k d_k^\times \sum_j s_{kj} B_j \rho_j^\times u_j \ddot{\theta}_j \right) \quad (6.162)$$

$$A_{0k} = B_k I_k u_k - \sum_j s_{jk} m_j d_j^\times B_k \rho_k^\times u_k \quad (6.163)$$

The right-hand-side term is

$$E_0 = T - \omega_0^\times h - \sum_k \left[(\dot{B}_k I_k \omega_k + B_k I_k \dot{B}_k^T \omega_0 \right. \quad (6.164)$$

$$\left. + m_k \left(\dot{d}_k^\times \omega_0^\times d_k + d_k^\times \omega_0^\times \dot{d}_k + d_k^\times \sum_j s_{kj} B_j (\Omega_j^\times \Omega_j^\times \rho_j) \right) \right] \quad (6.165)$$

where T is the external torque on the entire spacecraft.

$A_{01}^T = A_{10}$ and $a_{12} = a_{21}$ since the inertia matrix is symmetric.

$$A_{k0} = u_k^T [B_k I_k B_k^T - m_k (B_k \rho_k)^\times d_k^\times] \quad (6.166)$$

The scalar inertia terms for the appendages come from the expression

$$u_k^T (B_k I_k u_k \ddot{\theta}_k - m_k (B_k \rho_k)^\times \sum_j s_{kj} B_j \rho_j^\times u_j \ddot{\theta}_j) \quad (6.167)$$

$$a_{11} = u_1^T (B_1 I_1 u_1 - m_1 (B_1 \rho_1)^\times s_{11} B_1 \rho_1^\times u_1) \quad (6.168)$$

$$a_{12} = -m_1 s_{12} u_1^T (B_1 \rho_1)^\times B_2 \rho_2^\times u_2 \quad (6.169)$$

$$a_{22} = u_2^T (B_2 I_2 u_2 - m_2 (B_2 \rho_2)^\times s_{22} B_2 \rho_2^\times u_2) \quad (6.170)$$

$$a_{21} = -m_2 s_{21} u_2^T (B_2 \rho_2)^\times B_1 \rho_1^\times u_1 \quad (6.171)$$

Note that $m_1 s_{12} = m_2 s_{21}$. To show that $a_{12} = a_{21}$

$$u_1^T (B_1 \rho_1)^\times B_2 \rho_2^\times u_2 = ((B_1 \rho_1)^\times B_2 \rho_2^\times u_2)^T u_1 \quad (6.172)$$

$$= u_2^T ((B_1 \rho_1)^\times B_2 \rho_2^\times)^T u_1 \quad (6.173)$$

Center-of-mass movement causes inertial coupling between the appendages. This disappears if $\rho_1 = 0$ or $\rho_2 = 0$. The right-hand-side terms contain all the remaining terms:

$$\begin{aligned} e_k = & T_k - u_k^T [\omega_0^\times (B_k I_k \omega_k + m_k (B_k \rho_k)^\times (\omega_0^\times d_k + \dot{d}_k)) \\ & + \dot{B}_k I_k \omega_k + B_k I_k \dot{B}_k^T \omega_0 + m_k (\dot{B}_k \rho_k)^\times (\omega_0^\times d_k + \dot{d}_k) \\ & + m_k (B_k \rho_k)^\times (\omega_0^\times \dot{d}_k + \omega_0^\times \omega_0^\times d_k + 2\omega_0^\times \dot{d}_k)] \end{aligned} \quad (6.174)$$

Checking symmetry in the inertia matrix is a good way to check the derivation. Angular-momentum conservation is shown in Example 6.6.

This model should agree with the gyrostat model. Example 6.7 shows that the two models do agree. The hub model propagates the wheel angles. This permits the incorporation of angle-dependent friction or other disturbances.

6.11. N-body hub with wheels

Fig. 6.5 shows a model of the James Webb Space Telescope. There are six reaction wheels, assumed to be ideal rotors and two appendages that are assumed to have three-degrees-of-freedom joints with respect to the core spacecraft. In this case, the appendages represent masses of fuel that move as a rigid body. The three degrees-of-freedom allow pendular motion and motion about the pendulum axis.

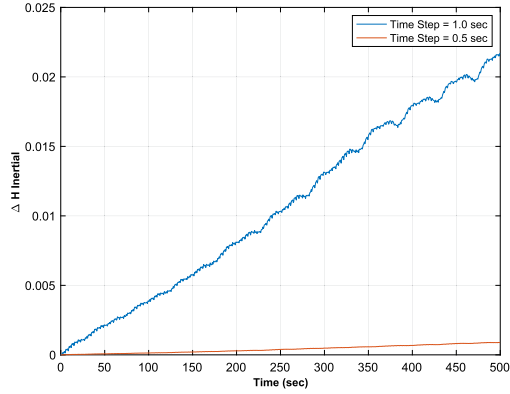
As a simplifying assumption, assume that the movement of fuel does not move the center-of-mass, that is $\dot{c} = 0$. The vector to the k th appendage body in the core frame and its derivatives are

$$d_k = \lambda_k + B_k \rho_k - c \quad (6.175)$$

```

1 d.inr = (diag([1 2 3]) diag([1 2 1]) diag
  ([0.2 2 0.5]));
2 d.m = [1 0.1 0.1];
3 d.u = [[0;0;0] [0;1;0] [1;0;0]];
4 d.rho = [[0;0;0] [0.1;0;0.3] [0;0;0.2]];
5 d.lambda = [[0;0;0] [0.1;0.1;0] [0;2;0.2]];
6 d.torque0 = [0;0;0]; d.torque = [0 0];
7 dT = [0.5 0.25]; n = 1000;
8 x = [1;0;0;0;.1;0.2;0.3;0;0;0.1;0.2];
9 h1 = zeros(1,n);
10 for k = 1:n
11     [~,h] = RHSNBodyCentralHub(x,0,d);
12     h1(k) = Mag(h);
13     x = RK4('RHSNBodyCentralHub',x,dT
  (1),0,d);
14 end
15 x = [1;0;0;0;.1;0.2;0.3;0;0;0.1;0.2];
16 h2 = zeros(1,2*n);
17 for k = 1:2*n
18     [~,h] = RHSNBodyCentralHub(x,0,d);
19     h2(k) = Mag(h);
20     x = RK4('RHSNBodyCentralHub',x,dT
  (2),0,d);
21 end
22 t1 = (0:(n-1))*dT(1); t2 = (0:(2*n-1))*dT(2);
23 NewFig('|H|');
24 plot(t1,abs(h1-h1(1)),t2,abs(h2-h2(1)));
25 legend('Time Step = 1.0 sec','Time Step = 0.5 sec');
26 xlabel('Time (sec)'); ylabel('\Delta H Inertial');
27 grid

```



Example 6.6: Angular-momentum conservation for a three-body model with a single degree-of-freedom hinges at the attachment points. The growth of momentum changes with the size of the time step.

$$\dot{d}_k = B_k \Omega_k^\times \rho_k \quad (6.176)$$

$$\ddot{d}_k = B_k \Omega_k^\times \Omega_k^\times \rho_k - B_k \rho_k^\times \dot{\Omega}_k \quad (6.177)$$

where B_k is the transformation matrix from the k th frame to the core frame. The total angular momentum in the core-body frame is

$$\begin{aligned}
 h = I_0 \omega_0 - m_0 d_0^\times d_0^\times + \sum_{k=1}^{N_w} u_k J_k (\Omega_k^w + u_k^T \omega_0) \\
 + \sum_{k=1}^{N_a} \left[B_k I_k (\Omega_k^s + B_k^T \omega_0) + m_k d_k^\times (\omega_0^\times d_k + \dot{d}_k) \right]
 \end{aligned} \quad (6.178)$$

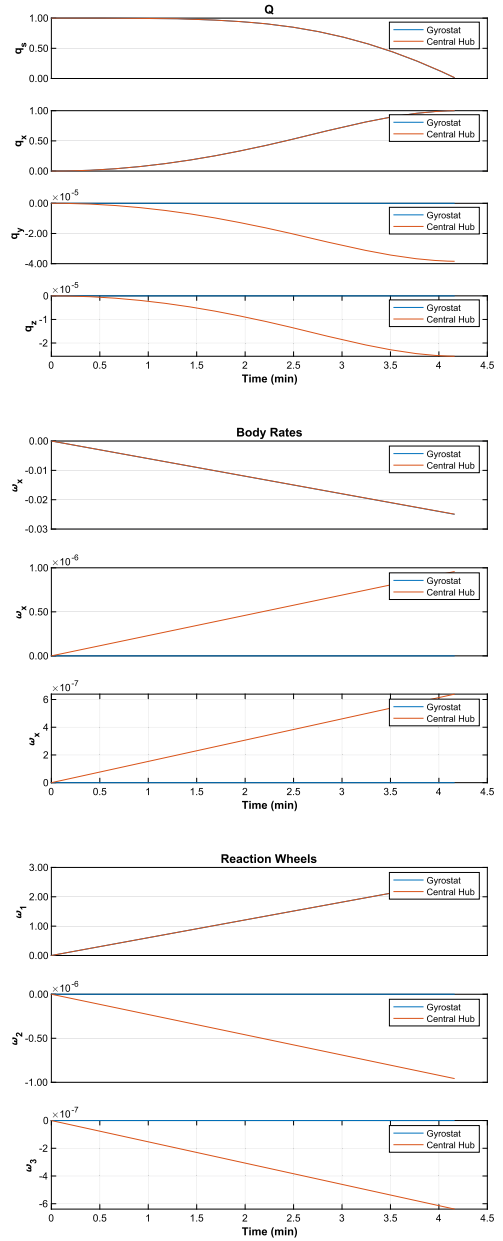
where J_k is the scalar reaction wheel inertia, u_k is the axis of rotation for the k th wheel, I_k is the appendage inertia, Ω_k is the appendage angular rate measured in the appendage frame, N_w is the number of wheels, and N_a is the number of appendages. Reorganizing

$$h = \left(I_0 - m_0 d_0^\times d_0^\times + \sum_{k=1}^{N_w} u_k J_k u_k^T + \sum_{k=1}^{N_a} [B_k I_k B_k^T - m_k d_k^\times d_k^\times] \right) \omega_0 \quad (6.179)$$

```

1 dCH.inr = {diag([1 2 3]) ...
2   diag([0.01 0 0]) diag([0 0.01 0]) diag([0 0
3     0.01])};
4 dCH.m = [1 0.1 0.1 0.1];
4 dCH.u = [0 1 0 0; 0 0 1 0; 0 0 0 1];
5 dCH.rho = zeros(3,4);
6 dCH.lambda = [0 0.1 0 0; 0 0 0.1 0; 0 0 0 0.1];
7 dCH.torque0 = [0;0;0];
8 dCH.torque = [0 0 0];
9
10 dGS = RHSGyrostat;
11 dGS.inr = dCH.inr{1};
12 dT = 0.25;
13 n = 1000;
14 xCH = [1;0;0;0; 0;0;0; 0;0;0; 0;0;0];
15 xGS = [1;0;0;0; 0;0;0; 0;0;0];
16 xPCH = zeros(13,n);
17 xPGS = zeros(10,n);
18 dCH.torque(1) = 0.0001;
19 dGS.torqueWheel(1) = 0.0001;
20 for k = 1:n
21   xPCH(:,k) = xCH;
22   xPGS(:,k) = xGS;
23   xCH = RK4('RHSNBodyCentralHub',xCH,dT,0,dCH);
24   xGS = RK4('RHSGyrostat',xGS,dT,0,dGS);
25 end
26
27
28 yL = {'q_s' 'q_x' 'q_y' 'q_z' ...
29       '\omega_x' '\omega_y' '\omega_z' ...
30       '\omega_1' '\omega_2' '\omega_3'};
31
32 t = (0:n-1)*dT;
33 l = {'Gyrostat' 'Central_Hub'};
34 k = 1:4;
35 pF = {[1 5] [2 6] [3 7] [4 8]};
36 TimeHistory(t,[xPGS(k,:);xPCH(k,:)],yL(k),'Q',pF
37   ,{1 1 1 1});
38
39 k = 5:7;
40 pF = {[1 4] [2 5] [3 6]};
41 TimeHistory(t,[xPGS(k,:);xPCH(k,:)],yL(k),'Body_
42   Rates',pF,{1 1 1});
43
44 k = 8:10;
45 kC = k+3;
46 pF = {[1 4] [2 5] [3 6]};
47 TimeHistory(t,[xPGS(k,:);xPCH(kC,:)],yL(k),'
48   Reaction_Wheels',pF,{1 1 1});

```



Example 6.7: Comparison of the gyrostat and hub models.

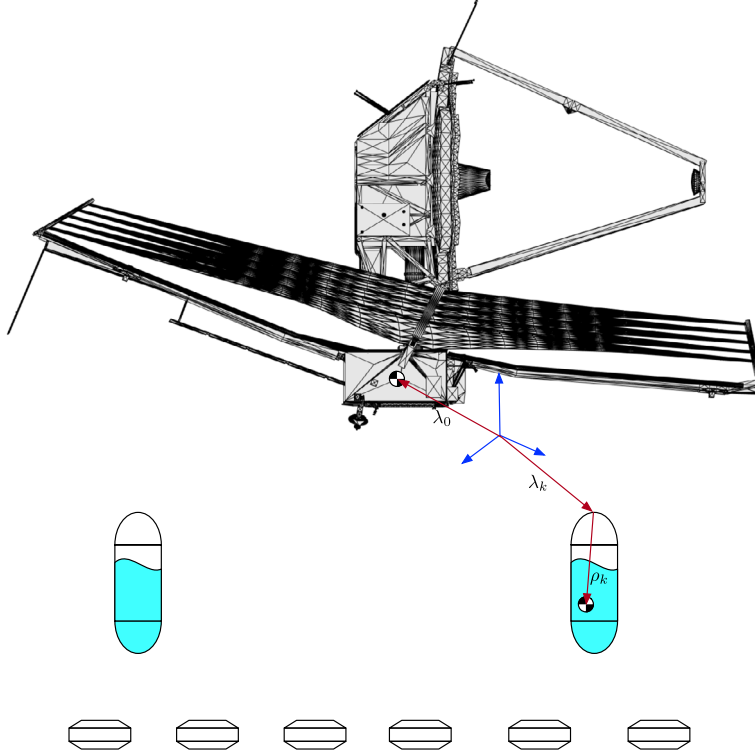


Figure 6.5 James Webb Space Telescope dynamical model. There is the core body, six reaction wheels assumed to be perfectly symmetric, and two fuel tanks.

$$+ \sum_{k=1}^{N_w} u_k J_k \Omega_k^w + \sum_{k=1}^{N_a} \left[B_k I_k \Omega_k + m_k d_k^\times \dot{d}_k \right]$$

The dynamical equation for the core is

$$T = \omega_0^\times h + \dot{h} \quad (6.180)$$

or

$$\begin{aligned} T = I \dot{\omega}_0 + & \left(\sum_{k=1}^{N_a} \left[\dot{B}_k I_k B_k^T + B_k I_k \dot{B}_k^T - m_k (\dot{d}_k^\times d_k^\times + d_k^\times \dot{d}_k^\times) \right] \right) \omega_0 \\ & + \sum_{k=1}^{N_w} u_k J_k \dot{\Omega}_k^w + \sum_{k=1}^{N_a} \left[\dot{B}_k I_k \Omega_k + B_k I_k \dot{\Omega}_k + m_k d_k^\times (B_k \Omega_k^\times \Omega_k^\times \rho_k - B_k \rho_k^\times \dot{\Omega}_k) \right] \\ & + \omega_0^\times h \end{aligned} \quad (6.181)$$

where

$$I = I_0 - m_0 d_0^\times d_0^\times + \sum_{k=1}^{N_w} u_k J_k u_k^T + \sum_{k=1}^{N_a} [B_k I_k B_k^T - m_k d_k^\times d_k^\times] \quad (6.182)$$

Rearranging to separate the inertia terms

$$\begin{aligned} T = & I \dot{\omega}_0 + \sum_{k=1}^{N_w} u_k J_k \dot{\Omega}_k^w + \sum_{k=1}^{N_a} (B_k I_k - m_k d_k^\times B_k \rho_k^\times) \dot{\Omega}_k \\ & + \sum_{k=1}^{N_a} \left[(\dot{B}_k I_k B_k^T + B_k I_k \dot{B}_k^T - m_k (\dot{d}_k^\times d_k^\times + d_k^\times \dot{d}_k^\times)) \omega_0 \right. \\ & + \dot{B}_k I_k \Omega_k + m_k d_k^\times B_k \Omega_k^\times \Omega_k^\times \rho_k \left. \right] \\ & + \omega_0^\times h \end{aligned} \quad (6.183)$$

These are the first three dynamical equations. The remaining equations are the reaction-wheel equations and the appendage equations. The reaction-wheel dynamical equations are

$$T_k = J_k (\dot{\Omega}_k + u_k \dot{\omega}_0) \quad (6.184)$$

The appendage dynamical equations are found by taking derivatives of the inertial angular momentum about the hinge.

$$H_k = AB_k I_k (\Omega_k + B_k^T \omega_0) + m_k (AB_k \rho_k)^\times \dot{D}_k \quad (6.185)$$

where

$$D_k = A d_k \quad (6.186)$$

$$\dot{D}_k = A (\omega_0^\times d_k + \dot{d}_k) \quad (6.187)$$

$$\ddot{D}_k = A \left(\omega_0^\times \omega_0^\times d_k + 2\omega_0^\times \dot{d}_k + \ddot{d}_k - d_k^\times \dot{\omega}_0 \right) \quad (6.188)$$

and

$$(AB_k \rho_k)^\times = AB_k \rho_k^\times B_k^T A^T \quad (6.189)$$

Collecting all the ω_0 terms

$$H_k = AB_k I_k (\Omega_k + B_k^T \omega_0) + m_k (AB_k \rho_k)^\times A (\dot{d}_k - d_k^\times \omega_0) \quad (6.190)$$

$$H_k = AB_k \left((I_k B_k^T - m_k \rho_k^\times B_k^T d_k^\times) \omega_0 + I_k \Omega_k + m_k \rho_k^\times B_k^T \dot{d}_k \right) \quad (6.191)$$

$$T_k = A^T B_k^T \dot{H}_k \quad (6.192)$$

Let

$$h_k = (I_k B_k^T - m_k \rho_k^\times B_k^T d_k^\times) \omega_0 + I_k \Omega_k + m_k \rho_k^\times B_k^T \dot{d}_k \quad (6.193)$$

The derivative of H_k is

$$\begin{aligned} T_k &= (B_k^T \omega_0^\times B_k + \Omega_k^\times) h_k \\ &+ (I_k \dot{B}_k^T - m_k \rho_k^\times \dot{B}_k^T d_k^\times - m_k \rho_k^\times B_k^T \dot{d}_k^\times) \omega_0 + m_k \rho_k^\times \dot{B}_k^T \dot{d}_k \\ &+ (I_k B_k^T - m_k \rho_k^\times B_k^T d_k^\times) \dot{\omega}_0 + I_k \dot{\Omega}_k \\ &+ m_k \rho_k^\times B_k^T (B_k \Omega_k^\times \Omega_k^\times \rho_k - B_k \rho_k^\times \dot{\Omega}_k) \end{aligned} \quad (6.194)$$

Rearranging to separate the inertia terms

$$\begin{aligned} T_k &= (I_k - \rho_k^\times \rho_k^\times) \dot{\Omega}_k + (I_k B_k^T - m_k \rho_k^\times B_k^T d_k^\times) \dot{\omega}_0 + (\omega_0^\times B_k + \Omega_k^\times) h_k \\ &+ (I_k \dot{B}_k^T - m_k \rho_k^\times \dot{B}_k^T d_k^\times - m_k \rho_k^\times B_k^T \dot{d}_k^\times) \omega_0 + m_k \rho_k^\times \dot{B}_k^T \dot{d}_k \\ &+ m_k \rho_k^\times \Omega_k^\times \Omega_k^\times \rho_k \end{aligned} \quad (6.195)$$

The generalized inertia matrix, for one wheel and one appendage, where body 2 is the wheel and body 3 is the appendage, is

$$G = \begin{bmatrix} I & u_k J_k & B_k I_k - m_k d_k^\times B_k \rho_k^\times \\ u_k^T J_k & J_k & 0 \\ I_k B_k^T - m_k \rho_k^\times B_k^T d_k^\times & 0 & I_k - m_k \rho_k^\times \rho_k^\times \end{bmatrix} \quad (6.196)$$

There is no inertial coupling between the appendages or the wheels. The (1,3) term should be the transpose of the (3,1) term.

$$\begin{aligned} (B_k I_k)^T &= I_k^T B_k^T = I_k B_k^T \\ (d_k^\times B_k \rho_k^\times)^T &= (B_k \rho_k^\times)^T (d_k^\times)^T = (\rho_k^\times)^T B_k^T (d_k^\times)^T = \rho_k^\times B_k^T d_k^\times \end{aligned} \quad (6.197)$$

which agree.

6.12. Control-moment gyros

This section derives the equations for a spacecraft with N single-axis control-moment gyros (CMGs). The CMG is shown in Fig. 6.6. The momentum wheel is assumed to be perfectly symmetric. The axis of gimbaling is aligned with its center-of-mass. Thus, rotating the CMG does not move the spacecraft's center-of-mass. CMGs should not be confused with spinning wheels on gimbaled platforms that are used for sensing.

The CMGs are numbered $k = 1$ through N . Each CMG has its base frame and its gimbaled frame. The base-to-spacecraft transformation matrix is B_{0k} . The gimbal-to-base transformation matrix is B_{gk} . B_{0k} is constant. Body 0 is the core spacecraft. The

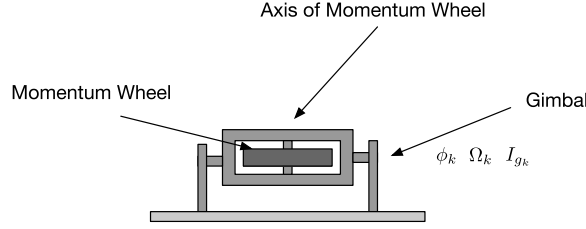


Figure 6.6 Single-gimbal control-moment gyro (CMG). The momentum wheel spins at a constant angular rate.

gimbal matrix for CMG k is

$$B_{gk} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \phi_k & -\sin \phi_k \\ 0 & \sin \phi_k & \cos \phi_k \end{bmatrix} \quad (6.198)$$

The gimbal axis is always the x -axis. The gimbal time derivative matrix is

$$\dot{B}_{gk} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & -\sin \phi_k & -\cos \phi_k \\ 0 & \cos \phi_k & -\sin \phi_k \end{bmatrix} \dot{\phi}_k \quad (6.199)$$

The external torque on the core spacecraft is T . The torque on gimbal k is T_{gk} . The torque on wheel k is T_{wk} . See Table 6.2 for all definitions of symbols used in the CMG derivation.

We write the total angular momentum in the inertial frame.

$$H = A \left(I_0 \omega + \sum_{k=1}^N B_{0k} \left[I_{gk} (\Omega_{gk} + B_{0k}^T \omega) + B_{gk} I_{wk} (\Omega_{wk} + B_{gk}^T B_{0k}^T \omega + B_{gk}^T \Omega_{gk}) \right] \right) \quad (6.200)$$

Let

$$h_k = I_{gk} (\Omega_{gk} + B_{0k}^T \omega) + B_{gk} I_{wk} (\Omega_{wk} + B_{gk}^T B_{0k}^T \omega + B_{gk}^T \Omega_{gk}) \quad (6.201)$$

Taking the derivative with respect to time and multiplying by A^T we get

$$T = I_0 \dot{\omega} + \omega^\times \left(I_0 \omega + \sum_{k=1}^N B_{0k} h_k \right) + \sum_{k=1}^N B_{0k} \dot{h}_k \quad (6.202)$$

and

$$\begin{aligned} \dot{h}_k = & I_{gk} (\dot{\Omega}_{gk} + B_{0k}^T \dot{\omega}) + \dot{B}_{gk} I_{wk} (\Omega_{wk} + B_{gk}^T B_{0k}^T \omega + B_{gk}^T \Omega_{gk}) \\ & + B_{gk} I_{wk} (\dot{\Omega}_{wk} + \dot{B}_{gk}^T B_{0k}^T \omega + B_{gk}^T B_{0k}^T \dot{\omega} + \dot{B}_{gk}^T \Omega_{gk} + B_{gk}^T \dot{\Omega}_{gk}) \end{aligned} \quad (6.203)$$

Table 6.2 CMG model symbols.

Symbol	Name	Units
B_{g_k}	Transformation matrix from the gimbal k to the base k frame	none
B_{0_k}	Transformation matrix from the CMG k base frame to the core 0 frame	none
T	External torque on the spacecraft	Nm
T_{g_k}	Torque on gimbal k	Nm
T_{w_k}	Torque on wheel k	Nm
I_0	Core spacecraft inertia	kg-m ²
I_{g_k}	Gimbal k inertia	kg-m ²
I_{w_k}	Wheel k inertia	kg-m ²
ϕ_k	Gimbal k angle	rad
Ω_{w_k}	Wheel k angular rate	rad/s
Ω_{g_k}	Gimbal k angular rate	rad/s
u_{g_k}	Unit vector along the gimbal rotational axis	none
u_{w_k}	Unit vector along the wheel rotational axis	none
A	Transformation matrix from the spacecraft body frame to the inertial frame	none
ω	Body rate in the body frame	rad/s

Note that

$$\dot{B}_{g_k} = B_{g_k} \Omega_{g_k}^\times \quad (6.204)$$

$$\dot{A} = A\omega^\times \quad (6.205)$$

The gimbal-torque equations are

$$T_{g_k} = u_{g_k} \dot{h}_k \quad (6.206)$$

The wheel-torque equations are

$$T_{w_k} = u_{w_k} I_{w_k} (\dot{\Omega}_{w_k} + \dot{B}_{g_k}^T B_{0_k}^T \omega + B_{g_k}^T B_{0_k}^T \dot{\omega} + \dot{B}_{g_k}^T \Omega_{g_k} + B_{g_k}^T \dot{\Omega}_{g_k}) \quad (6.207)$$

To which we add the kinematic equations

$$\dot{q} = f(q, \omega) \quad (6.208)$$

$$\dot{\phi}_k = \Omega_{g_k} \quad (6.209)$$

where q is the quaternion from the inertial frame to the spacecraft frame and u_k is the unit vector along the axis of rotation.

The dynamical equations need to be assembled in the form

$$T = I(\phi)\dot{\omega} + g(\omega, \phi) \quad (6.210)$$

where I is the generalized inertia matrix and, in this case, ω is a vector of all angular rates that is of length $3 + 2N$. We rearrange $\dot{h}_k$ to make the division clearer

$$\begin{aligned} \dot{h}_k = & \left(I_{g_k} B_{0_k}^T + B_{g_k} I_{w_k} B_{g_k}^T B_{0_k}^T \right) \dot{\omega} \\ & + (I_{g_k} + B_{g_k} I_{w_k} B_{g_k}^T) \dot{\Omega}_{g_k} + B_{g_k} I_{w_k} \dot{\Omega}_{w_k} \\ & + \dot{B}_{g_k} I_{w_k} (\Omega_{w_k} + B_{g_k}^T B_{0_k}^T \omega + \Omega_{g_k}) \\ & + B_{g_k} I_{w_k} \dot{B}_{g_k}^T (B_{0_k}^T \omega + \Omega_{g_k}) \end{aligned} \quad (6.211)$$

We assemble the inertia matrix from this equation. It should be symmetric. Looking at only one CMG we get

$$I = \begin{bmatrix} I_0 + B_{0_k} \left(I_{g_k} + B_{g_k} I_{w_k} B_{g_k}^T \right) B_{0_k}^T & B_{0_k} (I_{g_k} + B_{g_k} I_{w_k} B_{g_k}^T) u_{g_k} & B_{0_k} B_{g_k} I_{w_k} u_{w_k} \\ u_{g_k}^T \left(I_{g_k} + B_{g_k} I_{w_k} B_{g_k}^T \right) B_{0_k}^T & u_{g_k}^T (I_{g_k} + B_{g_k} I_{w_k} B_{g_k}^T) u_{g_k} & u_{g_k}^T B_{g_k} I_{w_k} u_{w_k} \\ u_{w_k}^T I_{w_k} B_{g_k}^T B_{0_k}^T & u_{w_k}^T I_{w_k} B_{g_k}^T u_{g_k} & u_{w_k}^T I_{w_k} u_{w_k} \end{bmatrix} \quad (6.212)$$

The inertia matrix is symmetric, as it should be. Note that there is no inertia coupling between the appendage and the reaction wheels. A simulation of a reaction-wheel torque pulse is shown in Example 6.8. The angular momentum is shown at two different timesteps. The number should be zero. It is lower when the time step is smaller, indicating that angular momentum is conserved. Both bodies react to the reaction-wheel torque. If we were modeling slosh, there would be a damper at the hinge of the second body.

6.13. Flexible structures

The simplest model of a flexible structure is shown in Fig. 6.7. The spacecraft is assumed to be composed of a rigid core to which a flexible structure is attached. The mass of the core is much larger than the mass of the structure, and the movement of the structure relative to the core is assumed to be small. Define

$$D = A(\lambda + r + \rho) \quad (6.213)$$

where the subscripts on r and ρ have been dropped. Taking derivatives

$$\dot{D} = A\omega^\times(\lambda + r + \rho) + A\dot{\rho} \quad (6.214)$$

$$\ddot{D} = A(\omega^\times\omega^\times(\lambda + r + \rho) + 2\omega^\times\dot{\rho} - (\lambda + r + \rho)^\times\dot{\omega} + \ddot{\rho}) \quad (6.215)$$

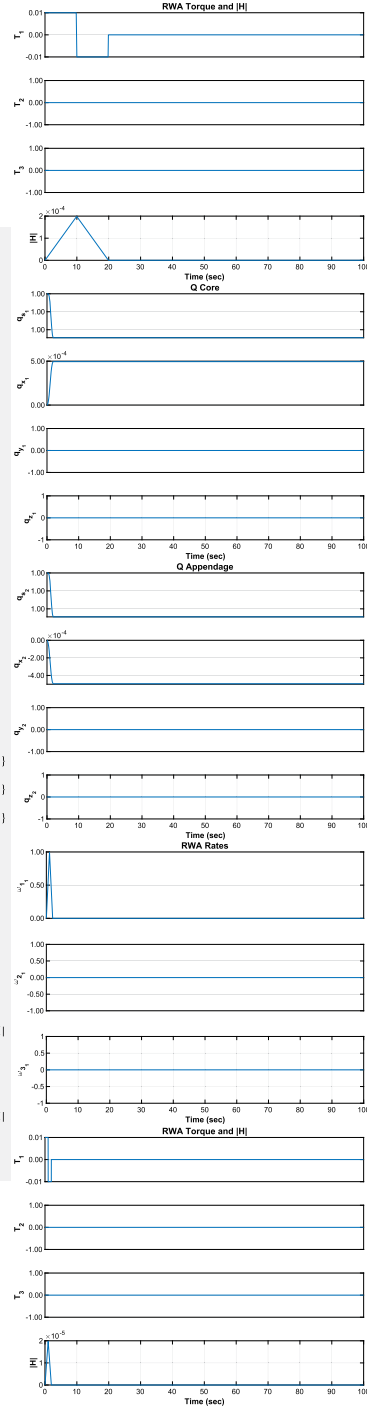
The total angular momentum of the system is

$$H = AI\omega + mD^\times\dot{D} \quad (6.216)$$

```

1 %% Checks two time steps
2 % Can be configured to have just reaction wheels
3 rWAOnly = false;
4 nPulse = 100;
5 dRHS = RHSGyroNBody;
6 tEnd = 100;
7 dT = 0.01;
8 q = [1;0;0;0];
9 if (rWAOnly)
10 dRHS.nB = 1;
11 x = [q;0;0;0;0;0;0];
12 m = 14;
13 else
14 dRHS.nB = [];
15 x = [q;q;0;0;0;0;0;0;0;0;0];
16 m = 21;
17 end
18 n = floor(tEnd/dT);
19 torqueW = [0,0,1;0;0]; % Internal torque
20 xP = zeros(m,n);
21 for k = 1:n
22 if (k < nPulse)
23 dRHS.torqueW = torqueW;
24 elseif (k > nPulse && k < 2*nPulse)
25 dRHS.torqueW = -torqueW;
26 else
27 dRHS.torqueW = [0;0;0];
28 end
29 [~,hX] = RHSGyroNBody(x,0,dRHS);
30 xP(:,k) = [x;dRHS.torqueW;Mag(hX)];
31 x = RK4('RHSGyroNBody',x,dT,0,dRHS);
32 end
33
34 t = (0:n-1)*dT;
35
36 yL = {'q_{s_1}' 'q_{x_1}' 'q_{y_1}' 'q_{z_1}' ...
37 'q_{s_2}' 'q_{x_2}' 'q_{y_2}' 'q_{z_2}' ...
38 '\omega_{x_1}' '\omega_{y_1}' '\omega_{x_1}' '\omega_{x_1}'
39 '\omega_{x_2}' '\omega_{y_2}' '\omega_{x_2}' '\omega_{x_2}'
40 '\omega_{y_1}' '\omega_{y_1}' '\omega_{y_1}' '\omega_{y_1}'
41 '\omega_{z_1}' '\omega_{z_1}' '\omega_{z_1}' '\omega_{z_1}'
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61 '\omega_{z_2}' '\omega_{z_2}' '\omega_{z_2}' '\omega_{z_2}'
62 Figui

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Example 6.8: Reaction-wheel pulse response for a spacecraft with two bodies and three reaction wheels.

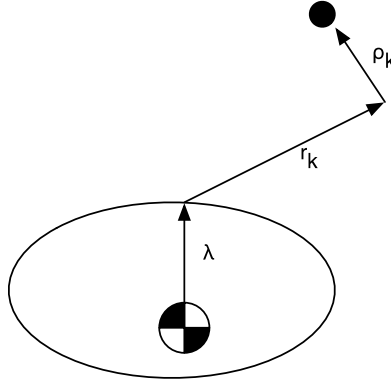


Figure 6.7 Flexible structure schematic.

where A transforms from the body to the inertial frame. The equations of motion of the entire spacecraft are

$$T = I\dot{\omega} + \omega^\times I\omega + m(\lambda + r + \rho)^\times (\omega^\times \omega^\times (\lambda + r + \rho) + 2\omega^\times \dot{\rho} - (\lambda + r + \rho)^\times \dot{\omega} + \ddot{\rho}) \quad (6.217)$$

If ρ is much smaller than r we can drop it from the $(\lambda + r + \rho)$ terms and keep only the derivative terms,

$$T = I\dot{\omega} + \omega^\times I\omega + m(\lambda + r)^\times (\omega^\times \omega^\times (\lambda + r) + 2\omega^\times \dot{\rho} - (\lambda + r)^\times \dot{\omega} + \ddot{\rho}) \quad (6.218)$$

Collecting terms,

$$T = I_T\dot{\omega} + \omega^\times I_T\omega + m(\lambda + r)^\times (2\omega^\times \dot{\rho} + \ddot{\rho}) \quad (6.219)$$

$$I_T = I - m(\lambda + r)^\times (\lambda + r)^\times \quad (6.220)$$

These equations are nonlinear in both body angular rate, ω , and mass displacement, ρ . The inertia matrix is constant because the flex deformation is assumed small.

The equation of motion for the mass is

$$F = m(\omega^\times \omega^\times (\lambda + r) + 2\omega^\times \dot{\rho} - (\lambda + r)^\times \dot{\omega} + \ddot{\rho}) \quad (6.221)$$

where F are all of the forces acting on the mass. The coupled equations are

$$\begin{bmatrix} I_T & m(\lambda + r)^\times \\ -m(\lambda + r)^\times & m \end{bmatrix} \begin{bmatrix} \dot{\omega} \\ \ddot{\rho} \end{bmatrix} = \begin{bmatrix} \omega^\times I_T\omega \\ m(\omega^\times \omega^\times (\lambda + r) + 2\omega^\times \dot{\rho}) \end{bmatrix} + \begin{bmatrix} T \\ f \end{bmatrix} \quad (6.222)$$

Linearization of these equations must be done with care. If both ω and $\dot{\rho}$ are small then they become

$$\begin{bmatrix} I_T & m(\lambda + r)^\times \\ -m(\lambda + r)^\times & m \end{bmatrix} \begin{bmatrix} \dot{\omega} \\ \ddot{\rho} \end{bmatrix} = \begin{bmatrix} T \\ f \end{bmatrix} \quad (6.223)$$

The coupling is entirely through the inertia matrix. If I_T is large and the rates are small the equations become

$$\begin{bmatrix} I_T & m(\lambda + r)^\times \\ -m(\lambda + r)^\times & m \end{bmatrix} \begin{bmatrix} \dot{\omega} \\ \ddot{\rho} \end{bmatrix} = \begin{bmatrix} \omega^\times I_T \omega \\ m\omega^\times \omega^\times (\lambda + r) \end{bmatrix} + \begin{bmatrix} T \\ f \end{bmatrix} \quad (6.224)$$

The mass still has a ω^2 term driving it but ρ only appears on the left-hand side. This derivation can be extended to multiple bodies. First, rewrite the equations in the form

$$T = I\dot{\omega} + \omega^\times I\omega - \sum_k (\lambda + r_k)^\times f_k \quad (6.225)$$

$$f_k = m(\omega^\times \omega^\times (\lambda + r_k) - (\lambda + r_k)^\times \dot{\omega} + \ddot{\rho}) \quad (6.226)$$

$$f_k = \sum_j k_{kj}(\rho_j - \rho_k) \quad (6.227)$$

where k_{kj} is the spring stiffness between j and k . Note that I is now just the core inertia. Just writing the equations for the masses

$$M\ddot{\rho} + K\rho = F \quad (6.228)$$

where K is symmetric and full. We define the following variables

$$\begin{aligned} \rho &= \Phi\eta \\ \Phi^T K \Phi &= \Omega^2 \\ \Phi^T M \Phi &= E \end{aligned}$$

where E is the identity matrix. Since ρ is a vector quantity, K is $3n$ -by- $3n$. If the number of modes is m , F is $3n$ by m . The mass equations become

$$\ddot{\eta} + \Omega^2 \eta = \Phi^T F \quad (6.229)$$

where F contains the force terms dependent on ω , Φ is known as the modal transformation matrix in finite-element models, η are the modal amplitudes, and Ω are the mode frequencies.

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